supermarket_sales.ipynb > empty cell

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```python
import pandas as pd          # Data handling
import numpy as np           # Numerical operations
import matplotlib.pyplot as plt  # Basic plotting
import seaborn as sns        # Advanced & beautiful plots
```

[55] ✓ 0.0s Python

1. **Load and Understand the Dataset**

```python
proj= pd.read_csv('supermarket_sales.csv')
```

[56] ✓ 0.0s Python

```python
proj.head()
```

[57] ✓ 0.0s Python

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	01-05-2019	1:08:00 PM	Ewallet	522.83	4.761905	26.1415	9.1
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	03-08-2019	10:29:00 AM	Cash	76.40	4.761905	3.8200	9.6
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155	340.5255	03-03-2019	1:23:00 PM	Credit card	324.31	4.761905	16.2155	7.4
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	8:33:00 PM	Ewallet	465.76	4.761905	23.2880	8.4
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085	634.3785	02-08-2019	10:37:00 AM	Ewallet	604.17	4.761905	30.2085	5.3

```python
proj.dtypes
```

[58] ✓ 0.0s Python

```
Invoice ID           object
Branch               object
City                 object
Customer type        object
Gender               object
Product line         object
Unit price          float64
Quantity              int64
Tax 5%              float64
```

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```
proj.dtypes
```

[58] ✓ 0.0s Python

```
Invoice ID                 object
Branch                     object
City                       object
Customer type              object
Gender                     object
Product line               object
Unit price                float64
Quantity                    int64
Tax 5%                    float64
Sales                     float64
Date                       object
Time                       object
Payment                    object
cogs                      float64
gross margin percentage   float64
gross income              float64
Rating                    float64
dtype: object
```

supermarket_sales.ipynb > empty cell

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2. **Data Cleaning**

```
proj.isnull()
```

[59] ✓ 0.0s Python

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
995	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
996	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
997	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
998	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
999	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False

1000 rows × 17 columns

```
proj.isnull().sum()
```

[60] ✓ 0.0s Python

```
Invoice ID       0
Branch           0
City             0
Customer type    0
Gender           0
Product line     0
Unit price       0
Quantity         0
Tax 5%           0
Sales            0
Date             0
Time             0
```

supermarket_sales.ipynb > empty cell

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```
Time                       0
Payment                    0
cogs                       0
gross margin percentage    0
gross income               0
Rating                     0
dtype: int64
```

[61]
```python
proj.duplicated()
```
✓ 0.0s Python

```
0      False
1      False
2      False
3      False
4      False
       ...
995    False
996    False
997    False
998    False
999    False
Length: 1000, dtype: bool
```

[62]
```python
proj.duplicated().sum()
```
✓ 0.0s Python

```
np.int64(0)
```

[63]
```python
proj
```
✓ 0.0s Python

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	01-05-2019	1:08:00 PM	Ewallet	522.83	4.761905	26.1415	9.1
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	03-08-2019	10:29:00 AM	Cash	76.40	4.761905	3.8200	9.6

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1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	03-08-2019	10:29:00 AM	Cash	76.40	4.761905	3.8200	9.6
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155	340.5255	03-03-2019	1:23:00 PM	Credit card	324.31	4.761905	16.2155	7.4
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	8:33:00 PM	Ewallet	465.76	4.761905	23.2880	8.4
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085	634.3785	02-08-2019	10:37:00 AM	Ewallet	604.17	4.761905	30.2085	5.3
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
995	233-67-5758	Giza	Naypyitaw	Normal	Male	Health and beauty	40.35	1	2.0175	42.3675	1/29/2019	1:46:00 PM	Ewallet	40.35	4.761905	2.0175	6.2
996	303-96-2227	Cairo	Mandalay	Normal	Female	Home and lifestyle	97.38	10	48.6900	1022.4900	03-02-2019	5:16:00 PM	Ewallet	973.80	4.761905	48.6900	4.4
997	727-02-1313	Alex	Yangon	Member	Male	Food and beverages	31.84	1	1.5920	33.4320	02-09-2019	1:22:00 PM	Cash	31.84	4.761905	1.5920	7.7
998	347-56-2442	Alex	Yangon	Normal	Male	Home and lifestyle	65.82	1	3.2910	69.1110	2/22/2019	3:33:00 PM	Cash	65.82	4.761905	3.2910	4.1
999	849-09-3807	Alex	Yangon	Member	Female	Fashion accessories	88.34	7	30.9190	649.2990	2/18/2019	1:28:00 PM	Cash	618.38	4.761905	30.9190	6.6

1000 rows × 17 columns

```python
proj.dtypes
```

[64] ✓ 0.0s Python

```
Invoice ID                object
Branch                    object
City                      object
Customer type             object
Gender                    object
Product line              object
Unit price               float64
Quantity                   int64
Tax 5%                   float64
Sales                    float64
Date                      object
Time                      object
```

supermarket_sales.ipynb > empty cell

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```
Time                        object
Payment                     object
cogs                       float64
gross margin percentage    float64
gross income               float64
Rating                     float64
dtype: object
```

```python
proj['Date'].dtype
```

[65] ✓ 0.0s Python

```
dtype('O')
```

```python
proj['Time'].dtype
```

[66] ✓ 0.0s Python

```
dtype('O')
```

```python
proj['Date'] = pd.to_datetime(proj['Date'], dayfirst=True, errors='coerce')
```

[67] ✓ 0.0s Python

```python
proj['Time'] = pd.to_datetime(proj['Time'], format='%I:%M:%S %p', errors='coerce').dt.time
```

[68] ✓ 0.0s Python

```python
proj.dtypes
```

[69] ✓ 0.0s Python

```
Invoice ID            object
Branch                object
City                  object
Customer type         object
```

Ln 1, Col 1 (17 selected) Spaces: 4 CRLF Cell 30 of 30

supermarket_sales.ipynb > empty cell

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```
Customer type                     object
Gender                            object
Product line                      object
Unit price                       float64
Quantity                           int64
Tax 5%                           float64
Sales                            float64
Date                      datetime64[ns]
Time                              object
Payment                           object
cogs                             float64
gross margin percentage          float64
gross income                     float64
Rating                           float64
dtype: object
```

3.DATA ANALYSIS USING NUMPY AND PANDAS

```python
total_sales = proj['Sales'].sum()
average_sales = proj['Sales'].mean()

print("Total Sales:", total_sales)
print("Average Sales:", average_sales)
```

[70] ✓ 0.0s Python

```
Total Sales: 322966.749
Average Sales: 322.966749
```

Total sales by product line:

```python
sales_by_product = proj.groupby('Product line')['Sales'].sum()
print(sales_by_product)
```

[71] ✓ 0.0s Python

```
Product line
Electronic accessories    54337.5315
```

supermarket_sales.ipynb > empty cell

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```
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages        56144.8440
Health and beauty         49193.7390
Home and lifestyle        53861.9130
Sports and travel         55122.8265
Name: Sales, dtype: float64
```

Average sales by gender:

```python
sales_by_gender = proj.groupby('Gender')['Sales'].mean()
print(sales_by_gender)
```

[72] ✓ 0.0s Python

```
Gender
Female    340.931414
Male      299.055738
Name: Sales, dtype: float64
```

Total sales by payment method:

```python
sales_by_payment = proj.groupby('Payment')['Sales'].sum()
print(sales_by_payment)
```

[73] ✓ 0.0s Python

```
Payment
Cash           112206.570
Credit card    100767.072
Ewallet        109993.107
Name: Sales, dtype: float64
```

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[74]

```python
plt.figure(figsize=(10,6))
sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
plt.title('Total Sales by Product Line')
plt.xticks(rotation=45)
```

supermarket_sales.ipynb > empty cell

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```
sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
plt.title('Total Sales by Product Line')
plt.xticks(rotation=45)
plt.ylabel('Total Sales')
plt.xlabel('Product Line')
plt.tight_layout()
plt.show()
```

[74] ✓ 0.1s

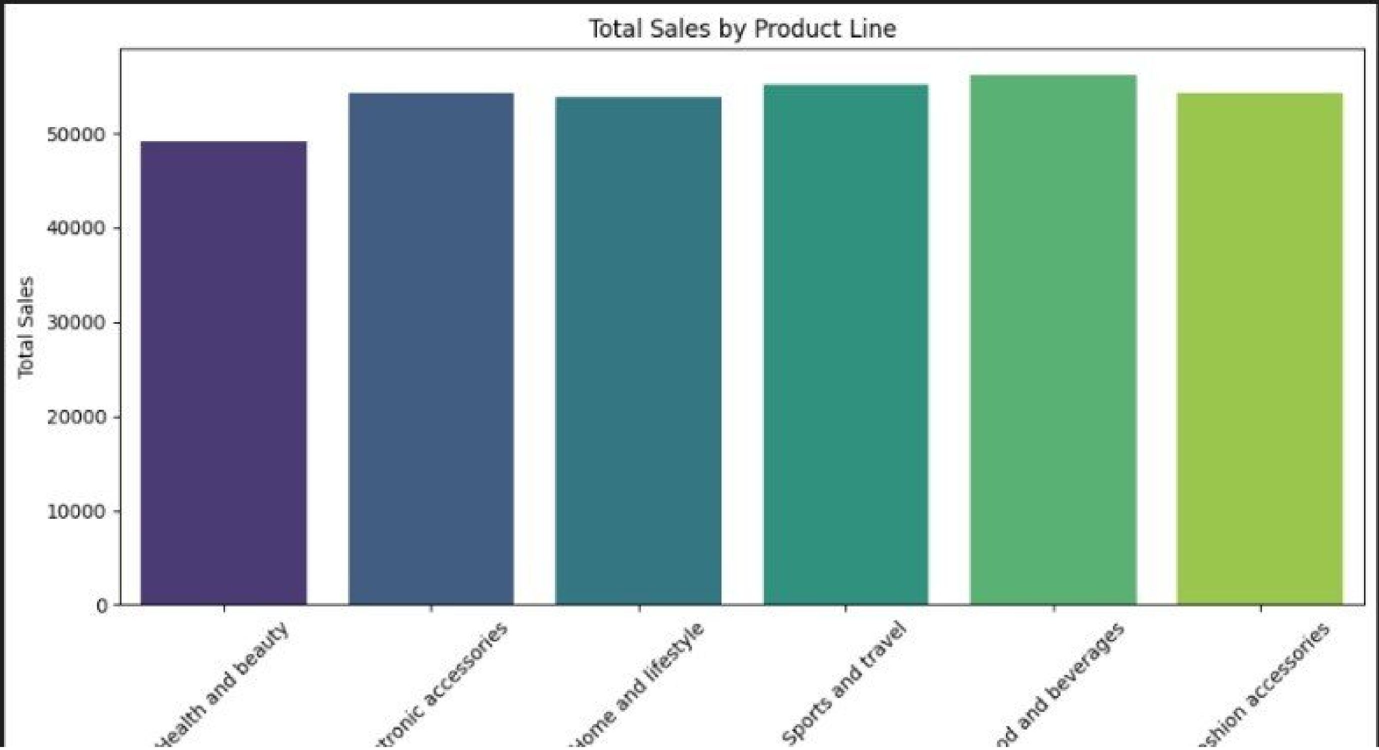
```
C:\Users\aditya\AppData\Local\Temp\ipykernel_16332\1540432024.py:2: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
C:\Users\aditya\AppData\Local\Temp\ipykernel_16332\1540432024.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
```



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```
sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
plt.title('Total Sales by Product Line')
plt.xticks(rotation=45)
plt.ylabel('Total Sales')
plt.xlabel('Product Line')
plt.tight_layout()
plt.show()
```

[74] ✓ 0.1s Python

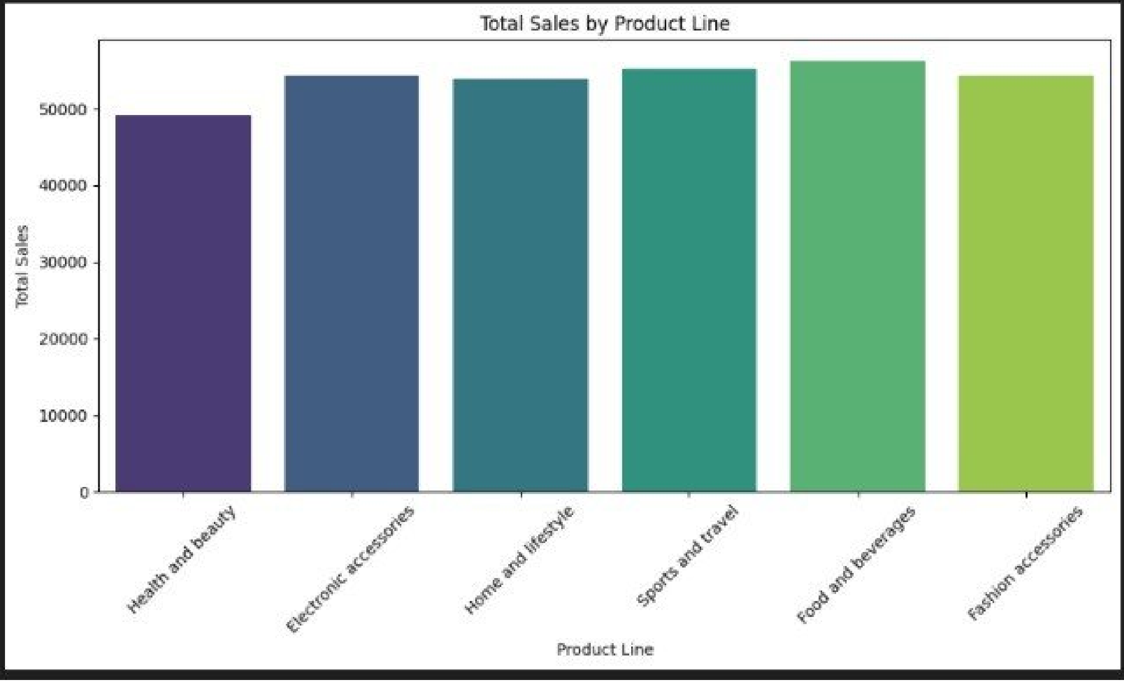
```
C:\Users\aditya\AppData\Local\Temp\ipykernel_16332\1540432024.py:2: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
C:\Users\aditya\AppData\Local\Temp\ipykernel_16332\1540432024.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x='Product line', y='Sales', data=proj, estimator=sum, ci=None, palette='viridis')
```



supermarket_sales.ipynb > empty cell

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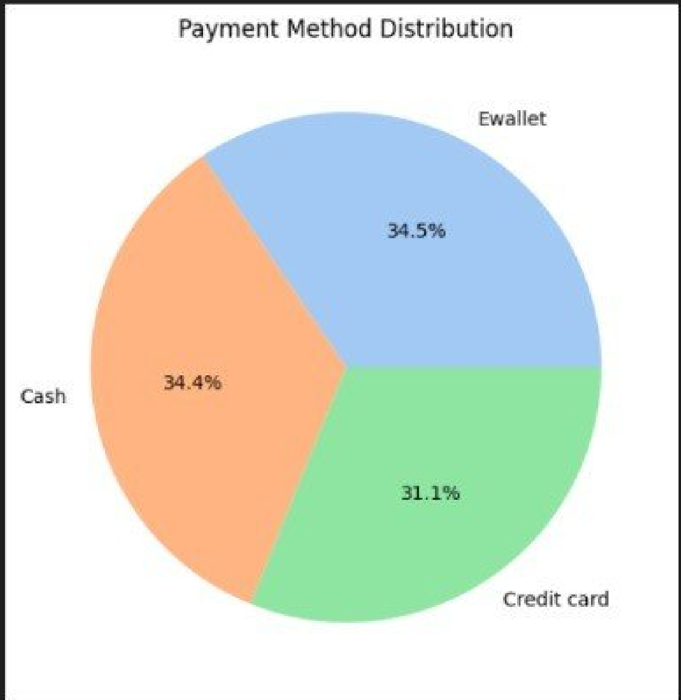
```python
payment_counts = proj['Payment'].value_counts()

plt.figure(figsize=(6,6))
plt.pie(payment_counts, labels=payment_counts.index, autopct='%1.1f%%', colors=sns.color_palette('pastel'))
plt.title('Payment Method Distribution')
plt.show()
```

[75] ✓ 0.0s Python



supermarket_sales.ipynb > empty cell

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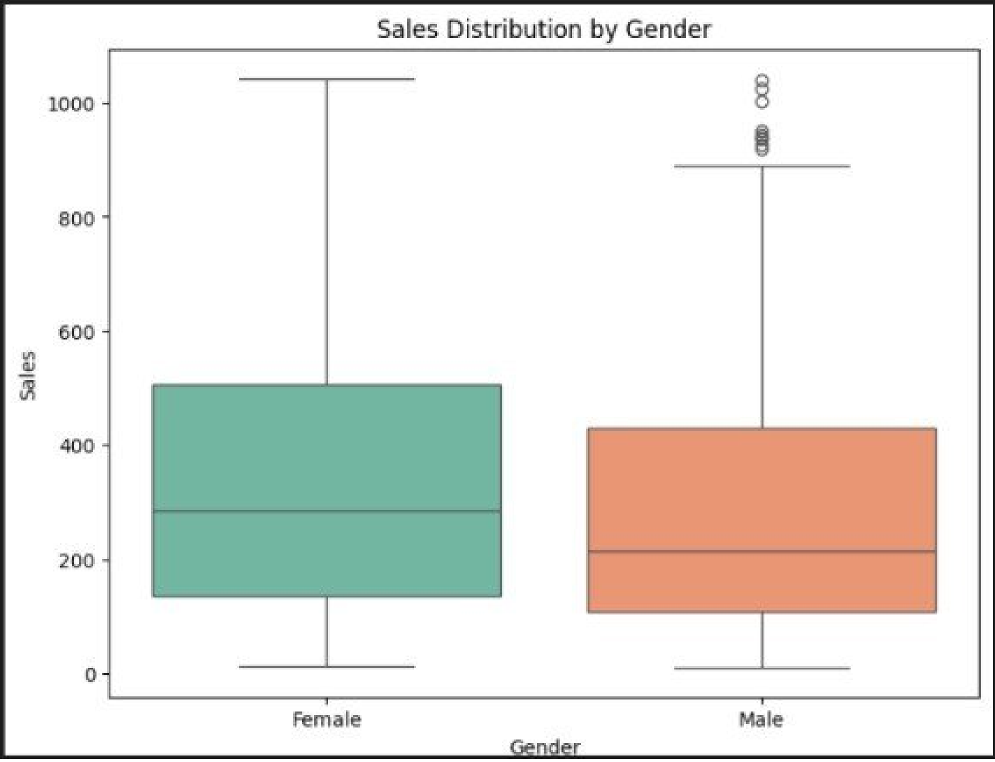
```python
plt.figure(figsize=(8,6))
sns.boxplot(x='Gender', y='Sales', data=proj, palette='Set2')
plt.title('Sales Distribution by Gender')
plt.xlabel('Gender')
plt.ylabel('Sales')
plt.show()
```

[76] ✓ 0.0s Python

```
C:\Users\aditya\AppData\Local\Temp\ipykernel_16332\3634443288.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(x='Gender', y='Sales', data=proj, palette='Set2')
```



supermarket_sales.ipynb > total_sales = proj['Sales'].sum()

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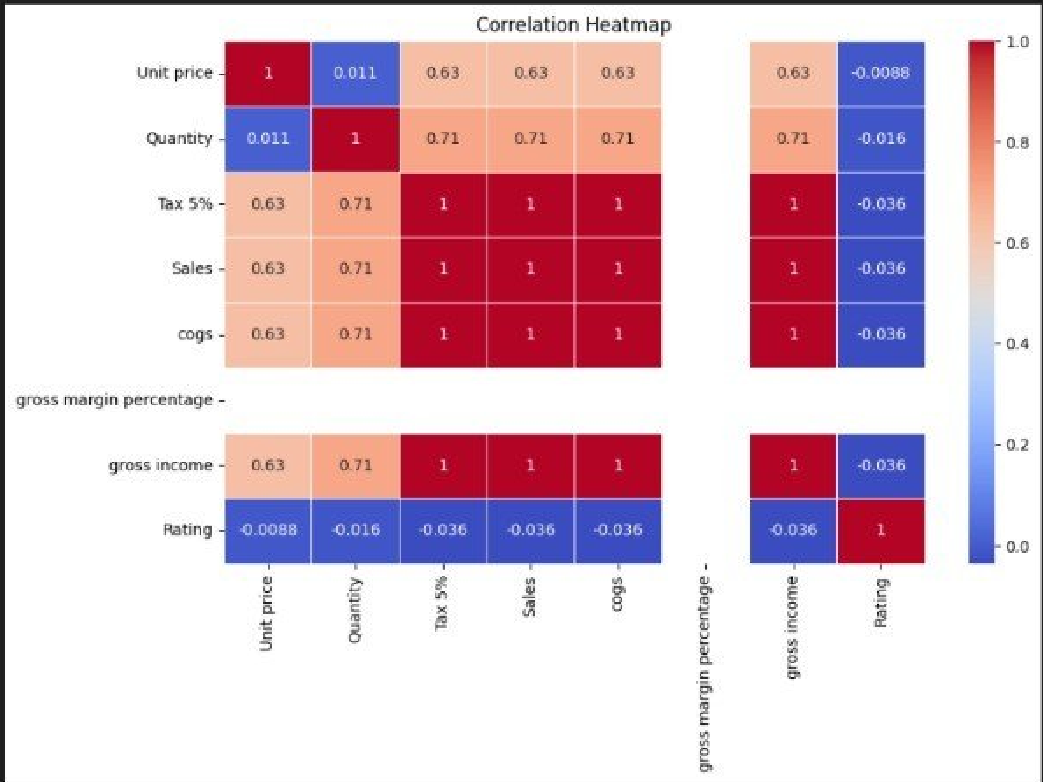
```python
plt.figure(figsize=(10,6))
corr = proj.select_dtypes(include=['float64', 'int64']).corr()

sns.heatmap(corr, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Heatmap')
plt.show()
```

[77] ✓ 0.1s Python



supermarket_sales.ipynb > PROJECTIVE OBJECTIVE:

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PROJECTIVE OBJECTIVE: -The goal of this project was to analyze a supermarket's sales data to understand customer behavior, sales trends, and how different product lines are performing.

DATA CLEANING: -Checked and handled missing values. -Removed duplicate rows. -Converted Date and Time columns into proper format. -Verified data types of all columns.

DATA ANALYSIS: -Calculated total and average sales. -Grouped the data by product line, gender, and payment method. -Found correlations between numerical columns.

VISUALISATION: -Bar plot to show total sales by product line. -Pie chart for payment method distribution. -Box plot to compare sales by gender. -Heatmap to see correlation between numeric features.

KEY OBSERVATION: -Food and Beverages had the highest total sales. -Most customers used Electronic Payment. -Sales distribution between Male and Female customers was nearly equal. -Quantity and Total had a strong positive correlation. -The dataset was mostly clean with very few missing values.

CREATED BY:

Aditya Ambara Matya

SUMMARY:

This project is about analyzing a supermarket sales dataset using Python. I used pandas and seaborn libraries to clean the data, check for missing values, and convert date and time columns. Then I analyzed things like which product line had the highest sales, what payment methods customers used the most, and how sales varied by gender. I also made some visualizations like bar charts, pie charts, and heatmaps to understand the data better. Overall, this helped me learn how to work with real-world data and find useful insights from it.